JAYPEE UNIVERSITY OF INFORMATION TECHNOLOGY
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DIGIMEDS : AN INTELLIGENT SYSTEM FOR DIGITIZING HANDWRITTEN MEDICAL PRESCRIPTIONS

GROUP NO.: 74

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MAJOR PROJECT - II (18B19CI891) | END TERM EVALUATION | ACADEMIC YEAR 2025-26 | May 14, 2026

1. PROBLEM STATEMENT

- Handwritten prescriptions are often illegible and misinterpreted.
- Leads to wrong medication, incorrect dosage, and treatment delays.
- Conventional OCR tools fail with varying handwriting and medical abbreviations.
- Lack of reliable digitization hinders creation of Electronic Health Records (EHRs).
- Hampers digital healthcare transformation in India.

2. OBJECTIVES

- Extract medicine name, dosage and frequency from handwritten prescriptions.
- Translate medical abbreviations into actionable schedules and reminders.
- Store prescription data digitally for easy access and history tracking.
- Provide medication reminders to improve adherence.
- Improve accuracy and reduce medication errors.

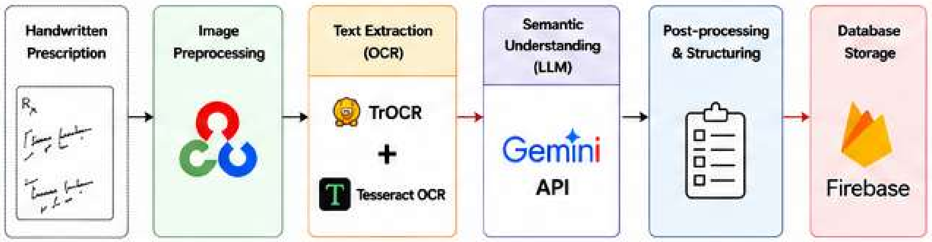
3. LITERATURE REVIEW & RESEARCH GAP

Paper / Authors	Method / Approach	Gap Identified
TrOCR OCR	Transformer based OCR	Moderate accuracy on cursive prescriptions
LayoutLMv2	Document AI + NLP	Focused on printed documents
MIRAGE Dataset	Generated medical prescription dataset	Synthetic data, limited real-world deployment
Recent MLLM Studies	Multimodal LLMs	High potential but lack end-to-end systems

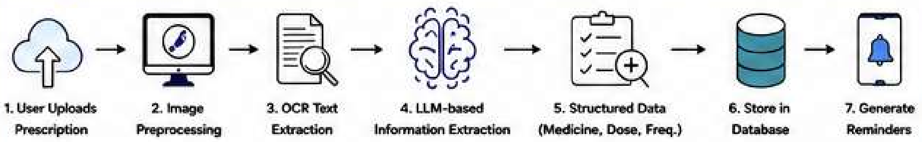
- Existing OCR struggles with high variability in handwritten prescriptions.
- Lack of medical context understanding for abbreviations and dosage.
- Limited systems offering real-time extraction + reminders + storage.

4. METHODOLOGY / APPROACH

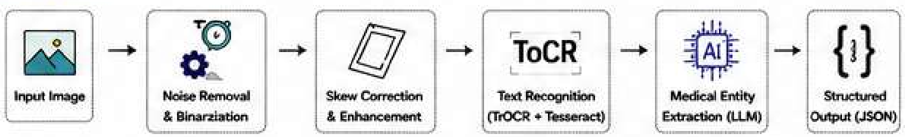
SYSTEM ARCHITECTURE



WORKFLOW DIAGRAM



OCR PIPELINE



5. IMPLEMENTATION DETAILS

TECHNOLOGIES USED

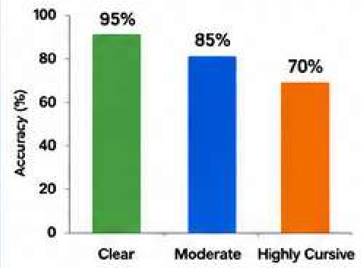
- Flutter (Mobile App)
- FastAPI (Backend API)
- OpenCV (Image Processing)
- Firebase (Database)
- Gemini API (LLM)
- Docker (Containerization)
- TrOCR (Text OCR)
- Python (Programming)

KEY SYSTEM MODULES

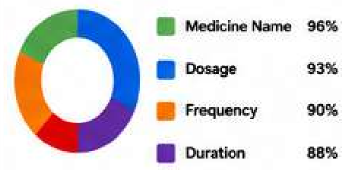
- **Image Upload Module** – Accepts prescription images
- **OCR & Extraction Module** – Extracts raw text from image
- **NLP & Understanding Module** – Extracts medicine, dosage, frequency
- **Scheduling & Reminder Module** – Converts to schedules and sends alerts
- **Database & History Module** – Stores data securely and manages history

6. RESULTS / PERFORMANCE

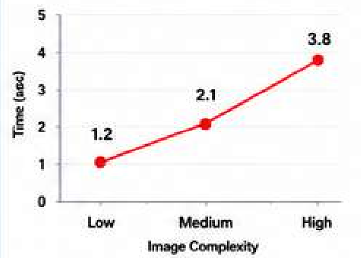
ACCURACY BY HANDWRITING QUALITY



FIELD-WISE EXTRACTION ACCURACY



AVERAGE RESPONSE TIME (SEC)



SAMPLE OUTPUT

Medicine	: Paracetamol 650mg
Dosage	: 1 Tablet
Frequency	: Twice a day
Duration	: 5 Days

Schedule Generated ✓

Reminder Set at: 9:00 AM & 9:00 PM

7. CONCLUSION

- DigiMeds successfully digitizes handwritten prescriptions using multimodal AI.
- High accuracy in extracting medicine, dosage and frequency.
- Automated scheduling and reminders improve medication adherence.
- Helps reduce medication errors and supports digital healthcare transformation.

8. FUTURE SCOPE

- Support for multi-language prescriptions.
- Drug interaction and allergy detection.
- Integration with Electronic Health Records (EHR) systems.
- Voice-based medication instructions.
- Cloud-based deployment for hospitals and clinics.

9. REFERENCES

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TECHNICAL SUMMARY

Platform	: Mobile + Web	Model Used	: TrOCR + Gemini API
Backend	: FastAPI (Python)	Deployment	: Dockerized Backend
Database	: Firebase (Cloud Firestore)	Frontend	: Flutter